HOLLAND BLUMER

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EXPERIENCE

Creative Coding NYC, New York — *Organizer*
Apr 2026 – Present

- Lead weekly meetups and drive community outreach.
- Coordinate event collaborations and sponsorships.
- Design promotional content using creative coding libraries.
- Liaise with community members contributing weekly posters.
- Manage social media and promote upcoming events.
- Volunteered at Processing Community Day NYC: speaker selection, open projector, promotion, and day-of support.

Recurse Center, Brooklyn — *Participant*
Mar 2026 – Jun 2026

- **Poster BluePrint**: An AI tool that identifies poster design elements that can be recreated with creative coding.
  - Built with Next.js/TypeScript + FastAPI/Python + GraphQL/Postgres/SQLAlchemy.
  - Matches design elements with OpenCV/SIFT image matching, using Google Gemini 2.5 Flash as a fallback.
  - Recreates matched elements with p5.js, SVG filters, Three.js, and GLSL shaders.
- **3D Motion Marbling**: An interactive experience that turns live hand movements into a virtual marbling comb.
  - Built with JavaScript + FastAPI/Python, using Google Cloud Storage and deployed on Render.
  - Tracks hands via webcam with MediaPipe, mapped to a MANO hand model, with OpenCV frame processing.
  - Displaces pixels on an HTML canvas in real time as the user moves.

Freelance, Remote — *Software Engineer & AI Trainer*
Jun 2022 – Present

- Lead end-to-end design and development for 12+ recurring clients in retail, real estate, restaurants, and private equity.
  - Translate client needs and brand identity into custom UX/UI through iterative wireframing and feedback.
  - Ship and maintain production sites.
- Run agentic AI coding workflows in containerized environments using OpenCode with open-source LLMs (GLM 4.7, Kimi K2-Thinking, MiniMax M2.5).
  - Evaluate generated solutions against test suites and identify specification gaps.
  - Write evaluation rubrics and review code patches to benchmark AI reasoning and implementation quality.
  - Work with Docker, Git, Python, JavaScript, and C.
- **Divot**: A B2C web app for sustainability, designed end to end.
  - Built with React/JavaScript + GraphQL + AWS (Amplify, EC2, Lambda, S3, DynamoDB, IAM) + Illustrator.

Hyundai Motor Group, Superior Township, MI — *Product Intelligence Intern*
Jun 2021 – Sep 2021

- Led creation of a 100+ page internal engineering guide for compact, midsize, and BEV pickup development, running bi-weekly meetings with engineering heads across 8 teams to manage progress.
- Traveled to HMG's Irvine, CA site to observe consumer interviews in test vehicles alongside a senior market research analyst, documenting feedback that informed EV feature planning.

EDUCATION

Dartmouth College — *M.Eng. Computer & Electrical Engineering*
2022

- **Helios**: A quality-control system for ChargePoint (capstone) capturing inspection images and production-line QA data.
  - Built with React/JavaScript + GraphQL + Python/SQL + AWS (EC2, Lambda, Amplify, S3, DynamoDB).
  - Includes a charger detail view, key manufacturing metrics, and an employee notes feature.
- **Relevant Coursework**: Machine Learning & Statistical Data Analysis, Smartphone Programming, Digital Image Processing, Computer Animation.

Northwestern University — *B.S. Manufacturing & Design Engineering, Minor in Data Science*
2021

- **UV Sense**: An NFC antenna for a L'Oréal-sponsored wearable UV sensor, later commercialized as My Skin Track UV.
  - Validated antenna connectivity through PCB prototypes built with electroplating, laser cutting, and soldering.